

A. Differential Encoding Architecture

Differential Encoding Pipeline

1. Reference Selection

Cryptographic random selection from reference genome pool

2. Chunking Strategy

Adaptive chunking based on analysis type (gene/window/variant)

3. Difference Computation

Compute variant differences: $\Delta = \text{Sample} - \text{Reference}$

4. Feature Extraction

Extract 384-dimensional feature vector

5. Hypervector Projection

Project to 10,000-D HDC space

6. Cryptographic Binding

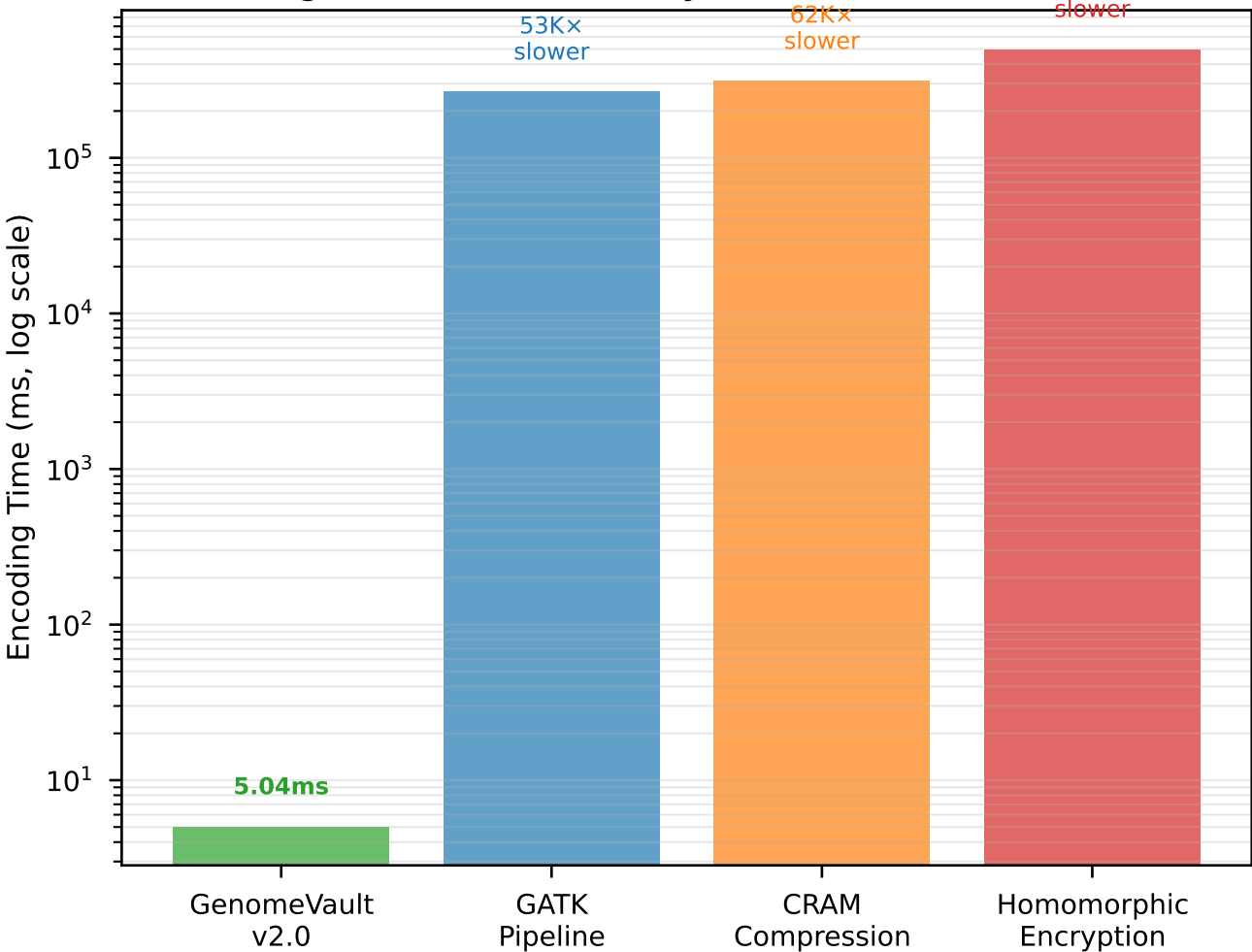
Bind with chunk identifiers and reference hash

7. Compression

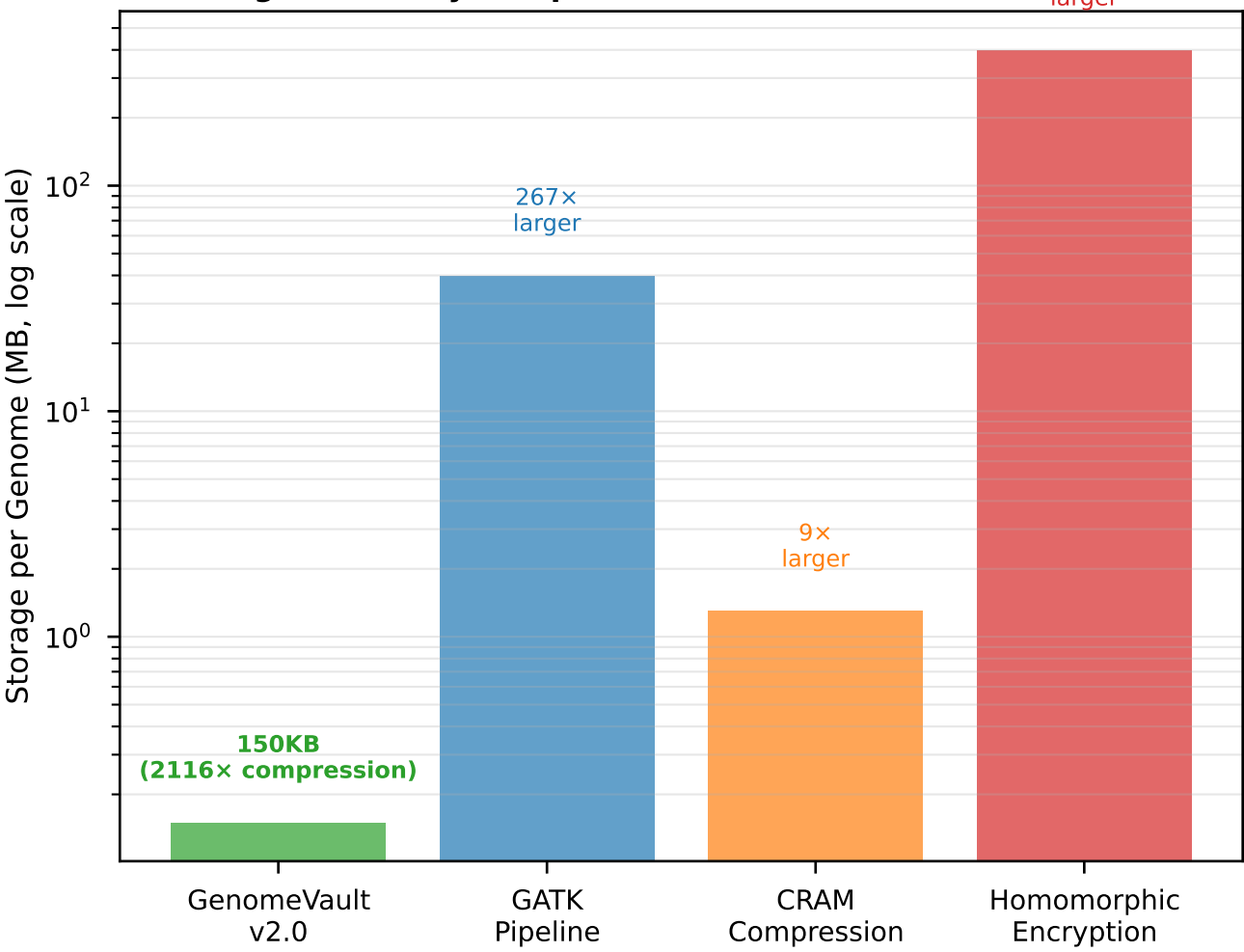
Apply sparsity and quantization

Result: 100-200 KB per genome with full reconstruction capability

B. Encoding Time vs Traditional Systems



C. Storage Efficiency Comparison



D. Processing Throughput

